Computer Studies Grade 12

Paper 1 – Revision Notes

Based on the 2025 Internal Examination Paper (ECZ/SC/2025/B5)

Topics Covered:

1. Hardware – Input/Output Devices and Memory
2. Operating Systems – Types, Interfaces and Scheduling
3. Data Representation – Analogue, Digital and Binary
4. Networks – Ring Topology
5. Logic Gates and Circuits
6. Applications Software – Presentations
7. Computers in Society – Hospitals and Robotics
8. Spreadsheets – Data Types and Formulas
9. Computer Security – Malware and Threats
10. Data Protection Act
11. Algorithms and Flowcharts
12. Systems Analysis and Design (SDLC)
13. Programming in C++
14. Databases and SQL

Topic 1: Hardware – Input/Output Devices and Memory

1.1 Multifunctional Devices

A multifunctional device (MFD) is a single piece of hardware that combines the functions of several separate devices.

Examples of Multifunctional Devices

- All-in-One printer (prints, scans, photocopies and faxes)
- Smartphone (phone, camera, calculator, GPS, media player)
- Tablet computer

> 📝 **Exam Question:** Give one example of a multifunctional device. [1]
> ☑ **Answer:** All-in-One printer / Smartphone

> ⚠ **Exam Tip:** Any device that combines at least two functions counts. State the device clearly.

1.2 Primary Memory (RAM and ROM)

Primary memory is directly accessible by the CPU. It stores data and instructions currently in use.

Type	Description
RAM (Random Access Memory)	Temporary, volatile memory. Stores data and programs currently being used. Lost when power is off.
ROM (Read Only Memory)	Permanent, non-volatile memory. Stores the BIOS/boot instructions. Cannot normally be written to by the user.
Cache Memory	Very fast memory between the CPU and RAM. Stores frequently used instructions to speed up processing.

> 📝 **Exam Question:** Name one example of primary memory. [1]
> ☑ **Answer:** RAM (Random Access Memory) / ROM (Read Only Memory) / Cache Memory

> ⚠ **Exam Tip:** Do not confuse primary memory (RAM/ROM) with secondary/backing storage (hard drive, USB).

1.3 Magnetic Ink Character Recognition (MICR)

MICR is used to read characters printed in magnetic ink, mainly on bank cheques.

Information Read by MICR on a Cheque

- Bank sort code (branch code)
- Account number
- Cheque number
- Amount printed at the bottom (added after processing)

Exam Question: List two pieces of information that are read on a cheque using MICR. [2]

Answer: Bank sort code AND Account number (or Cheque number)

⚠ **Exam Tip:** Remember: MICR reads the pre-printed numbers at the bottom of a cheque. These are in a special magnetic ink font.

Topic 2: Operating Systems

2.1 Types of Operating Systems

OS Type	Description / Example
Single-user, Single-tasking	Allows one user to run one program at a time. e.g. MS-DOS
Single-user, Multi-tasking	One user runs multiple programs simultaneously. e.g. Windows, macOS
Multi-user	Multiple users share system resources simultaneously. e.g. Linux server, UNIX
Real-Time OS (RTOS)	Processes data as it arrives with minimal delay. e.g. embedded systems, medical devices
Batch Processing OS	Groups similar jobs and processes them together without user interaction. e.g. payroll systems
Network OS	Manages network resources and allows computers to communicate. e.g. Windows Server

Exam Question: State two types of operating systems. [2]

Answer: Single-user / Multi-user / Real-Time OS / Batch Processing OS (any two)

2.2 Types of User Interfaces

Interface Type	Description
GUI (Graphical User Interface)	Uses icons, windows, menus and a pointer/mouse. e.g. Windows desktop
CLI (Command Line Interface)	User types text commands. e.g. MS-DOS prompt, Linux terminal
Menu-Driven Interface	User selects from a list of options. e.g. ATM screen, restaurant ordering kiosk
Voice/Natural Language Interface	User interacts using spoken or typed natural language. e.g. Siri, Cortana
Touch Interface	User interacts by touching the screen. e.g. smartphone, tablet

> 📝 **Exam Question:** List two types of user interfaces. [2]
> ☑ **Answer:** GUI (Graphical User Interface) AND CLI (Command Line Interface) (any two)

2.3 Scheduling

Scheduling is the method by which the operating system decides the order in which processes are given access to the CPU.

Why Scheduling is Important

- Ensures the CPU is used efficiently
- Allows multiple processes to share the CPU fairly
- Prevents any one process from monopolising the CPU
- Minimises waiting time for users

Common Scheduling Algorithms

- Round Robin – each process gets an equal time slice in rotation
- First Come First Served (FCFS) – processes are executed in arrival order
- Priority Scheduling – higher-priority processes run first
- Shortest Job First (SJF) – shortest process is selected next

> 📝 **Exam Question:** Explain the meaning of the term "Scheduling". [2]
> ☑ **Answer:** Scheduling is the process by which the operating system decides the order and amount of time each process/task is allocated to use the CPU, ensuring efficient use of processor time.

> ⚠ **Exam Tip:** For 2 marks you need a definition plus a purpose. Mention the OS, the CPU, and the idea of managing/ordering processes.

Topic 3: Data Representation – Analogue, Digital and Binary

3.1 Analogue vs Digital Signals

Feature	Analogue Signal vs Digital Signal
Nature	Analogue: Continuous, smooth wave that varies infinitely | Digital: Discrete, takes only two values (0 or 1)
Representation	Analogue: Represented by a sine wave | Digital: Represented as square pulses (on/off)
Examples	Analogue: Sound waves, temperature, old radio | Digital: Computer data, CDs, digital TV
Noise affected?	Analogue: Easily affected by noise/interference | Digital: Less affected; errors easier to detect

> 📝 **Exam Question:** Explain the difference between an analogue signal and a digital signal. [4]
> ☑ **Answer:** An analogue signal is a continuous signal that varies smoothly and can take any value within a range (e.g. sound waves), whereas a digital signal is a discrete signal that can only take two

values – on (1) or off (0) – represented as square pulses. Analogue signals are more susceptible to noise and interference than digital signals.

3.2 Binary Conversion – Denary to Binary

Method: Repeated Division by 2

Divide the denary number repeatedly by 2. Record the remainders from bottom to top.

Worked Example: Convert 50 to Binary

Step-by-step division:

Division	Remainder
50 ÷ 2 = 25	0
25 ÷ 2 = 12	1
12 ÷ 2 = 6	0
6 ÷ 2 = 3	0
3 ÷ 2 = 1	1
1 ÷ 2 = 0	1

Reading remainders from bottom to top: $50_{10} = 110010_2$

Verification using Place Values

32 + 16 + 2 = 50 ✓

Bit Position (Power of 2)	Value
Bit 5 ($2^5 = 32$)	1 × 32 = 32
Bit 4 ($2^4 = 16$)	1 × 16 = 16
Bit 3 ($2^3 = 8$)	0 × 8 = 0
Bit 2 ($2^2 = 4$)	0 × 4 = 0
Bit 1 ($2^1 = 2$)	1 × 2 = 2
Bit 0 ($2^0 = 1$)	0 × 1 = 0

> **Exam Question:** Convert the denary number 50 into binary. [2]
> **Answer:** 110010

> ⚠ **Exam Tip:** Always show your working (the division table). Even if your answer is wrong, you may get a mark for the method. Verify by adding up the place values.

Topic 4: Networks – Ring Topology

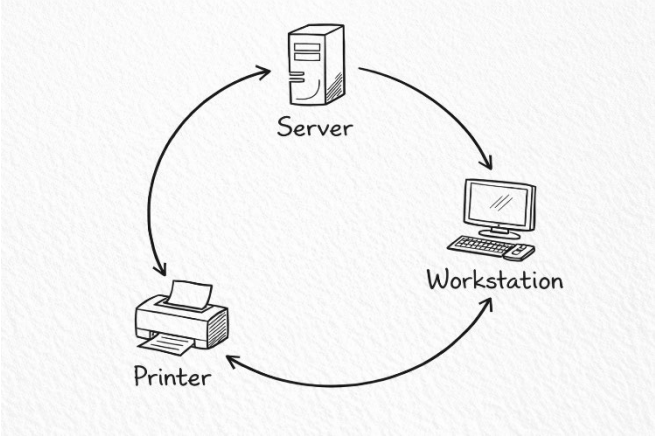
4.1 Ring Topology

In a Ring Topology, all devices (nodes) are connected in a closed loop. Data travels around the ring in one direction from node to node until it reaches its destination.

Diagram Description (for exam drawings)

- Draw devices in a circular arrangement
- Connect each device to the next with a line forming a complete circle
- Label: Server, Workstation, Printer (or any combination of devices)
- Show data flow arrows around the ring (unidirectional)

📝 **Exam Question:** Draw and label a Ring Topology Network containing a server, a workstation and a printer. [3]



☑ **Answer:** Draw three devices in a circle, each connected to the next. Label: Server → Workstation → Printer → (back to Server). Show directional arrows.

⚠ **Exam Tip:** You get marks for: the ring/circle arrangement, correct labelling of at least the three named devices, and connecting lines. Always use a ruler and label clearly.

4.2 Functions of a Server

- Stores and manages shared files and data
- Controls access to the network and authenticates users
- Manages shared resources such as printers and internet connection
- Provides email and communication services
- Performs backups of network data

📝 **Exam Question:** State one function of a server in the network. [1]

☑ **Answer:** A server stores and manages shared files/resources and controls user access to the network.

4.3 Advantages and Disadvantages of Ring Topology

Advantages	Disadvantages
Data flows in one direction, reducing collisions	If one device or cable fails, the entire network goes down
All devices have equal access to the network	Adding or removing devices disrupts the network
Performs well under heavy traffic conditions	Fault diagnosis is difficult
No need for a central hub or switch	Slower than star topology for small networks

> **Exam Question:** Give one drawback of a Ring Topology Network. [1]
> **Answer:** If one device or cable fails, the entire network is disrupted / breaks down.

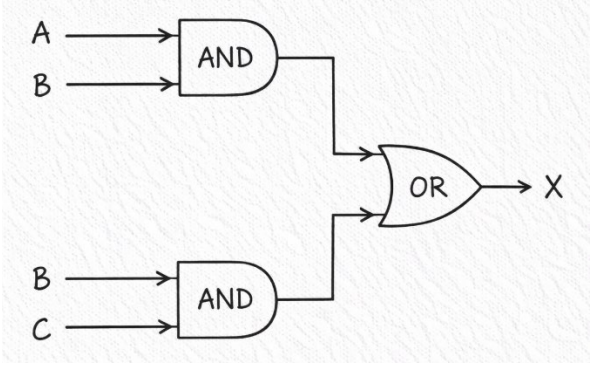
Topic 5: Logic Gates and Circuits

5.1 Basic Logic Gates

Gate	Symbol Description and Rule
AND	Output is 1 ONLY when BOTH inputs are 1. Symbol: D-shape with flat left side.
OR	Output is 1 when AT LEAST ONE input is 1. Symbol: D-shape with curved left side.
NOT	Inverts the input. Output is 1 when input is 0 (and vice versa). Symbol: Triangle with a circle.
NAND	Output is 0 ONLY when BOTH inputs are 1 (NOT AND). AND gate with a circle at output.
NOR	Output is 0 when ANY input is 1 (NOT OR). OR gate with a circle at output.

5.2 Logic Expression: X = (A AND B) OR (B AND C)

Steps to draw this circuit:



1. Draw an AND gate with inputs A and B. Label output AB.
2. Draw a second AND gate with inputs B and C. Label output BC.
3. Feed both AND gate outputs (AB and BC) into an OR gate.
4. The output of the OR gate is X.

Truth Table for X = (A AND B) OR (B AND C)

A	B	C	A AND B	B AND C	X (result)
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	1	1
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	1	0	1
1	1	1	1	1	1

> **Exam Question:** Name one home appliance that uses logic circuits. [1]
>
> **Answer:** Washing machine / Microwave oven / Dishwasher / Refrigerator / Air conditioner

> ⚠ **Exam Tip:** Logic circuits are used in any appliance with electronic controls. Mention a common household appliance with a digital control panel.

Topic 6: Applications Software – Presentations

6.1 Presentation Software

Presentation software is used to create slide-based visual presentations combining text, images, charts and media.

Examples of Presentation Software

- Microsoft PowerPoint (most common)
- Google Slides
- LibreOffice Impress
- Prezi

> **Exam Question:** Name an application she could use to prepare a report as a presentation. [1]
>
> **Answer:** Microsoft PowerPoint

6.2 Devices for Presenting

- Projector (connected to a laptop/computer)
- Interactive Whiteboard (Smart Board)
- Large screen monitor / TV display
- Tablet connected to a screen

Exam Question: Give one device she could use to present her report. [1]
Answer: A projector (to display the presentation on a screen for the audience)

6.3 Types of Data Entered in Slide Placeholders

- Text (headings, bullet points, descriptions)
- Images / photographs
- Charts and graphs
- Videos and audio clips
- Tables

Exam Question: Give one type of data that should be entered in the placeholder on the slide. [1]
Answer: Text / Image / Chart (any one)

Topic 7: Computers in Society

7.1 Computers in Hospitals

Benefits	Disadvantages
Patient records stored digitally – easy to access and share	High cost of equipment and maintenance
Faster and more accurate diagnosis using imaging/scanning	Risk of data loss if system fails or is hacked
Computer-controlled equipment increases precision in surgery	Staff need training to use the systems
Appointment scheduling and billing are automated	Patient data privacy concerns (hacking, data breaches)
Drug dispensing systems reduce medication errors	Over-reliance on computers – problems if system goes down

Exam Question: Give one benefit of using computers in hospitals. [1]
Answer: Patient records can be stored and accessed quickly and easily / Computer-controlled equipment increases precision in diagnoses and surgery.

Exam Question: State one disadvantage of using computers in hospitals. [1]
Answer: Risk of patient data being hacked or lost / High cost of purchasing and maintaining computer systems.

7.2 Robots in Manufacturing

Tasks Robots Perform in a Factory

- Welding and cutting of materials
- Painting and surface finishing of furniture parts
- Lifting and moving heavy pieces of wood/furniture
- Assembly – fitting parts together accurately
- Quality control – checking measurements and defects using sensors
- Packing and labelling finished products

Effects of Robots on Workers

- Job losses (redundancy) – robots replace human workers
- Workers may be redeployed to supervise and maintain robots
- Workers may need retraining in technical skills
- Improved safety – dangerous tasks are done by robots

> **Exam Question:** State two tasks a robot would do in the factory. [2]
>
> **Answer:** Welding / Painting / Assembling / Lifting and moving heavy materials (any two)

> **Exam Question:** Give one effect the use of robots would have on workers of the company. [1]
>
> **Answer:** Workers may lose their jobs (be made redundant) as robots replace them.

Topic 8: Spreadsheets

8.1 Data Types in Spreadsheet Cells

Data Type	Description and Examples
Text (Label)	Alphabetic or alphanumeric characters. Left-aligned by default. e.g. "Name", "Chongololo", "Class 12A"
Number (Value)	Numeric data used in calculations. Right-aligned by default. e.g. 45, 3.5, 100
Date/Time	Stored as numbers but displayed as dates. e.g. 01/01/2025
Formula	An expression that performs a calculation. Always starts with =. e.g. =SUM(B2:D2)
Boolean	Logical TRUE or FALSE value

> **Exam Question:** State the type of data entered in cell A3. [1]
>
> **Answer:** Text (Label) – cell A3 likely contains a class name or student name which is alphanumeric/text data.

8.2 Common Spreadsheet Formulas

Formula	What it Does
=SUM(B3:D3)	Adds all values from cell B3 to D3

Formula	What it Does
=AVERAGE(B3:D3)	Calculates the average of B3 to D3
=COUNT(B3:D3)	Counts the number of cells with values
=MAX(B3:D3)	Returns the largest value
=MIN(B3:D3)	Returns the smallest value
=IF(condition, value_if_true, value_if_false)	Returns one value or another based on a condition

9 The following spreadsheet shows part of a results analysis done by a school.

	A	B	C	D	E
1			RESULT ANALYSIS		
2	Class	Passed	Failed	Average	Total
3	12N	45	5		
4	12J	40	10		
5	12A	47	3		
6	12s	49	1		

Exam Question: Write a formula that can be used to calculate the total number of learners in class in cell E3. [2]

Answer: =SUM(B3:D3) OR =B3+C3+D3 (depending on which cells contain the learner counts)

⚠ **Exam Tip:** In the exam, read the spreadsheet carefully to identify which cells to include in the SUM. Always start formulas with the = sign.

Topic 9: Computer Security – Malware and Threats

9.1 Types of Malware

Type	Description
Virus	A program that attaches itself to files and spreads when files are shared. Can corrupt or delete data.
Worm	Self-replicating program that spreads over networks without user action.
Trojan Horse	Disguises itself as legitimate software but performs harmful actions when run.
Spyware	Secretly monitors user activity and sends data to a third party.
Ransomware	Encrypts the victim's files and demands payment for the decryption key.

Type	Description
Adware	Displays unwanted advertisements; may also track browsing habits.
Rootkit	Gives unauthorised users administrator-level access while hiding itself.

> 📝 **Exam Question:** Give one example of malware. [1]
> ☑ **Answer:** Virus / Worm / Trojan Horse / Spyware / Ransomware (any one)

9.2 Methods of Protecting Against Malware

- Install and regularly update antivirus/anti-malware software
- Keep the operating system and all software up to date with security patches
- Use a firewall to monitor and filter incoming and outgoing network traffic
- Avoid downloading files or clicking links from unknown sources
- Use strong, unique passwords and enable two-factor authentication
- Regularly back up important data to an external drive or cloud
- Encrypt sensitive data so it is unreadable if stolen

> 📝 **Exam Question:** Describe two methods that can be used to protect a computer system from malware. [2]
> ☑ **Answer:** 1. Install antivirus software that scans and removes malware. 2. Keep the operating system updated with security patches to close known vulnerabilities.

9.3 Physical Threats to Computer Systems

- Fire – can destroy hardware and stored data
- Flooding / water damage
- Theft of hardware
- Power surges or sudden power failures
- Vandalism – deliberate physical damage to equipment
- Natural disasters (earthquakes, lightning strikes)

> 📝 **Exam Question:** State one physical threat to computer systems. [1]
> ☑ **Answer:** Fire / Theft / Flooding / Power surge / Vandalism (any one)

> ⚠ **Exam Tip:** Physical threats are things that can physically damage or remove the hardware. Do not confuse with logical/software threats like viruses.

Topic 10: Data Protection Act

10.1 Importance of the Data Protection Act

- Protects individuals from misuse of their personal data held by organisations
- Gives individuals the right to know what data is held about them

- Ensures data is kept secure and not accessed by unauthorised persons
- Prevents personal data from being sold or shared without consent
- Holds organisations legally responsible for how they handle data

> **Exam Question:** Explain the importance of the Data Protection Act. [2]
>
> **Answer:** The Data Protection Act is important because it protects individuals' personal information from being misused, accessed without permission, or shared without consent. It gives people rights over their own data and makes organisations legally responsible for keeping data secure.

10.2 Principles of the Data Protection Act

5. Data must be collected and processed fairly and lawfully
6. Data must be obtained for specified, explicit and legitimate purposes
7. Data must be adequate, relevant and not excessive for its purpose
8. Data must be accurate and kept up to date
9. Data must not be kept longer than necessary
10. Data must be processed in accordance with the rights of the data subject
11. Appropriate security measures must be taken to protect data
12. Data must not be transferred to countries without adequate protection

> **Exam Question:** State two principles of the Data Protection Act. [2]
>
> **Answer:** 1. Data must be kept secure and protected from unauthorised access. 2. Data must not be kept longer than necessary / Data must be accurate and up to date. (Any two from the list above)

Topic 11: Algorithms and Flowcharts

11.1 Flowchart Symbols

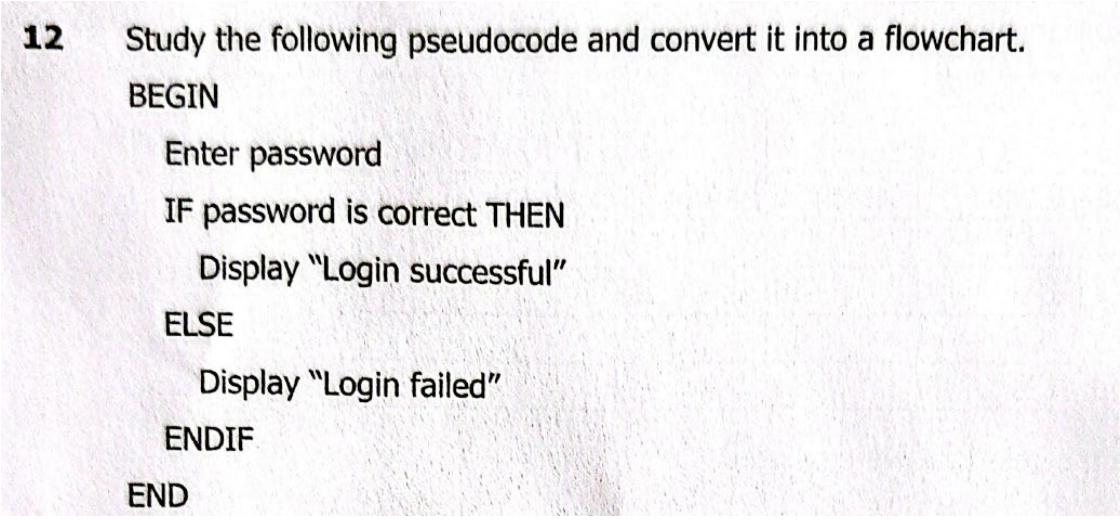
Symbol (Shape)	Meaning / Use
Oval / Rounded Rectangle	START or END of the algorithm
Parallelogram	INPUT or OUTPUT operation
Rectangle	PROCESS – a calculation or action
Diamond	DECISION – a question with YES/NO or TRUE/FALSE branches
Arrow	Shows the flow/direction of the algorithm

11.2 Converting Pseudocode to a Flowchart

The following pseudocode from the exam must be converted to a flowchart:

```
BEGIN
  Enter password
```

```
  IF password is correct THEN
    Display "Login successful"
  ELSE
    Display "Login failed"
END
```

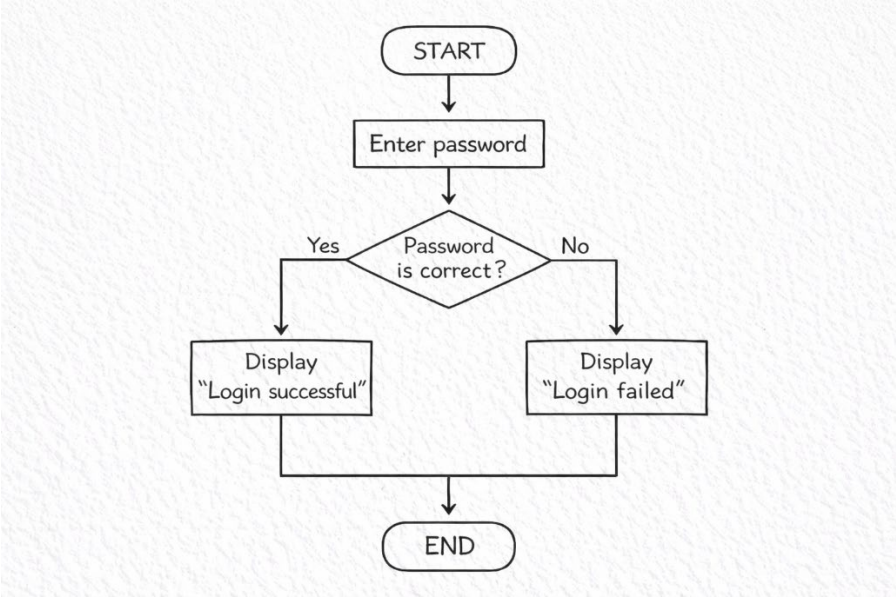


12 Study the following pseudocode and convert it into a flowchart.

BEGIN
Enter password
IF password is correct THEN
Display "Login successful"
ELSE
Display "Login failed"
ENDIF
END

Step-by-step flowchart description:

13. OVAL: START
14. PARALLELOGRAM: INPUT Password
15. DIAMOND: Is password correct? (YES / NO branches)
16. YES branch → PARALLELOGRAM: OUTPUT "Login successful"
17. NO branch → PARALLELOGRAM: OUTPUT "Login failed"
18. Both branches meet → OVAL: END



> **Exam Question:** Study the pseudocode and convert it into a flowchart. [4]
>
> **Answer:** Draw: START → INPUT Password → DECISION (correct?) → YES: OUTPUT "Login successful" / NO: OUTPUT "Login failed" → END

⚠ **Exam Tip:** Marks are awarded for: correct use of shapes (1), correct START/END (1), correct decision diamond with two branches (1), correct output statements (1). Always connect all paths back to END.

Topic 12: Systems Analysis and Design (SDLC)

12.1 Systems Development Life Cycle (SDLC) Stages

Stage	Key Activities
1. Feasibility Study	Assess if the project is worth doing (cost, time, technical resources)
2. Investigation / Analysis	Gather information on the current system. Identify problems and user requirements.
3. Design	Design inputs, outputs, file structures, processes, user interface and security.
4. Implementation (Programming)	Write, test and debug the programs. Create documentation.
5. Testing	Test the system thoroughly using test data (normal, extreme, erroneous).
6. Installation / Conversion	Install the new system and convert data from old to new format.
7. Maintenance	Correct errors, update and improve the system after installation.

12.2 Converting Paper Files to Electronic Format

- Scanning documents using a scanner (creates digital image files)
- Optical Character Recognition (OCR) to convert scanned text into editable digital text
- Manual data entry – typing data directly into the computer
- Using a barcode reader or MICR to capture data from printed documents

📝 **Exam Question:** State two ways the Systems Analyst could use to convert paper-based files into electronic format. [2]

☑ **Answer:** 1. Scanning the documents and using OCR software. 2. Manual data entry / keyboard data entry.

12.3 Computer-Based vs Manual System

Advantage of Computer-Based System	Drawback of Computer-Based System
Faster processing and retrieval of information	High initial cost of hardware and software

Advantage of Computer-Based System	Drawback of Computer-Based System
Data can be stored more compactly and securely	Risk of data loss due to hardware failure or hacking
Easier to update, sort and search records	Staff require training; learning curve
Multiple users can access data simultaneously	Power dependency – system unusable during power cuts

12.4 Methods of Gathering Information (Investigation Stage)

- Interviews – talking directly to users of the current system
- Questionnaires – written questions distributed to users
- Observation – watching how the current system is used
- Document inspection – examining existing forms, reports and files
- Research – studying similar systems elsewhere

> **Exam Question:** State one method the Systems Analyst could use to get information on the manual filing system. [1]
>
> **Answer:** Interview / Questionnaire / Observation / Document inspection (any one)

12.5 Design Stage Activities

- Designing input forms and output reports
- Designing the file/database structure (tables, fields, data types)
- Designing the user interface (screens and menus)
- Designing the system architecture and hardware requirements
- Creating data flow diagrams (DFDs) and entity relationship diagrams (ERDs)

> **Exam Question:** Name two activities the Systems Analyst should carry out during the design stage. [2]
>
> **Answer:** 1. Designing the input forms and output reports. 2. Designing the database/file structure (fields and data types).

12.6 System Documentation

Type of Documentation	Purpose
Technical/System Documentation	For maintenance programmers. Includes program code, system architecture, data structures, test plans, DFDs.

Type of Documentation	Purpose
User Documentation	For end users. Explains how to use the software (menus, input, output). Includes user manual / help guide.

> **Exam Question:** State the documentation the Systems Analyst would provide for maintenance of the system. [1]
>
> **Answer:** Technical documentation (also called system/maintenance documentation) – includes program code, system architecture, data flow diagrams, test results and file structures.

12.7 Methods of Implementing a New System

Method	Description
Direct (Immediate) Changeover	Old system is stopped and new system is started all at once. Risky but cheap.
Parallel Running	Old and new systems run simultaneously for a period. Safe but expensive (double the work).
Phased Implementation	New system is introduced in stages/modules one at a time.
Pilot Implementation	New system is introduced to one department/site first before full rollout.

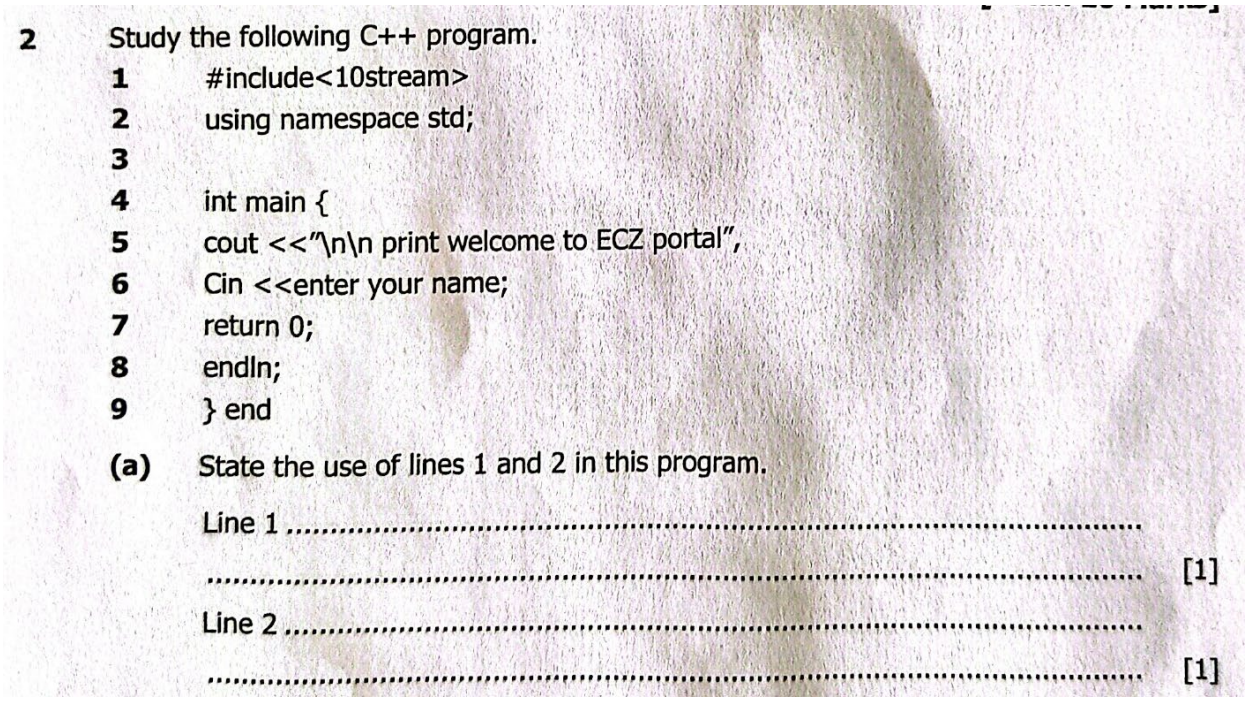
> **Exam Question:** Mention two methods of implementing the computer-based system. [2]
>
> **Answer:** Direct changeover / Parallel running / Phased implementation / Pilot implementation (any two)

Topic 13: Programming in C++

13.1 Basic C++ Program Structure

The exam presents the following C++ program:

```
Line 1:  #include<iostream>
Line 2:  using namespace std;
Line 3:  (blank)
Line 4:  int main {                    ← ERROR
Line 5:    cout << "print the code";
Line 6:    cin << enter your name;     ← ERROR
Line 7:    return 0;
Line 8:  }
```



2 Study the following C++ program.

```
1   #include<10stream>
2   using namespace std;
3
4   int main {
5   cout <<"\n\n print welcome to ECZ portal",
6   Cin <<enter your name;
7   return 0;
8   endln;
9   } end
```

(a) State the use of lines 1 and 2 in this program.

Line 1 ..... [1]

Line 2 ..... [1]

13.2 Lines 1 and 2 Explained

#include<iostream>: A preprocessor directive that includes the input/output stream library, enabling the use of cin (input) and cout (output) functions.

using namespace std;: Tells the compiler to use the standard (std) namespace, so you can write cout and cin instead of std::cout and std::cin.

> **Exam Question:** State the use of lines 1 and 2 in this program. [2]
>
> **Answer:** Line 1 (#include<iostream>): includes the input/output library so cin and cout can be used. Line 2 (using namespace std): allows use of standard C++ functions without the std:: prefix.

13.3 Debugging the Errors

Line	Error and Correction
Line 4: int main {	Error: Missing parentheses after main. Corrected: int main() {
Line 6: cin << enter your name;	Two errors: (1) cin uses >> not <<. (2) The string must be in double quotes and stored in a variable. Corrected: cin >> name; (assuming string name; is declared)

> **Exam Question:** Identify the errors on lines 4 and 6 and provide the corrected code. [4]
>
> **Answer:** Line 4: int main { → Corrected: int main() { (parentheses added after main). Line 6: cin << enter your name; → Corrected: cin >> name; (changed << to >>, used a variable, added proper string handling).

13.4 Keywords in C++

A keyword (reserved word) in C++ is a word that has a special, predefined meaning in the language. It cannot be used as a variable name or identifier.

Examples of C++ Keywords

- int, float, double, char, bool – data types
- if, else, while, for, do – control flow
- return, void, main – function-related
- cout, cin – input/output objects (from the std library)
- include, using, namespace, class, public, private

> **Exam Question:** Explain the meaning of the term "keyword" in C++. [2]
>
> **Answer:** A keyword (also called a reserved word) is a word that has a special predefined meaning in C++ and is part of the programming language syntax. It cannot be used as a variable name or identifier. Examples include int, if, else, while, return.

Topic 14: Databases and SQL

14.1 Database Concepts

Term	Definition
Table	A structured collection of related data organised in rows and columns.
Record (Row)	One complete set of data about a single entity (one row in the table).
Field (Column)	A single attribute/characteristic of an entity (one column in the table).
Primary Key	A field that uniquely identifies each record in a table. No duplicates; cannot be null.
Foreign Key	A field in one table that refers to the primary key of another table (creates a link/relationship).

14.2 The Database from the Exam

The school fees database contains: Student_No, Last_name, First_name, Gender, School_fees.

Field	Example Value
Student_No (Primary Key)	NJ001, NJ004, NJ008
Last_name	Student surname
First_name	Luyando, Ronald
Gender	M / F

Field	Example Value
School_fees	1 500.00, 3 000.00

📝 **Exam Question:** Give one reason why Student_No has been selected as a primary key. [1]

3 The following database shows part of records of student fees kept by a school accounts office.

Student _No	Last_name	First_name	Gender	School fees (ZMW)
N3001	Banda	Mwaka	F	2 000.00
NJ004	Phiri	Luyando	F	1 000.00
NJ008	Nyambe	Ronald	M	1 500.00
NJ201	Chowe	Nchimunya	M	3 000.00

☑ **Answer:** Student_No uniquely identifies each student – no two students have the same number, so it can be used to distinguish each record from all others.

📝 **Exam Question:** State the number of records in this database. [1]

☑ **Answer:** 3 records (NJ001, NJ004, NJ008 – count the data rows, not the header row)

⚠ **Exam Tip:** A record = one row of data. The header row (field names) does NOT count as a record.

14.3 SQL – INSERT Statement

Syntax:

```
INSERT INTO table_name (field1, field2, ...) VALUES (value1, value2, ...);
```

📝 **Exam Question:** Write an SQL statement to insert the following record: Student_no: NJ401, Lastname: Bwalya, Firstname: Bupe, Gender: F, School_fees: 2500.00 [3]

```
INSERT INTO Students (Student_No, Last_name, First_name, Gender,
School_fees) VALUES ('NJ401', 'Bwalya', 'Bupe', 'F', 2500.00);
```

(c) Write an SQL statement that could be used to insert the following record in the database.

Student_no: NJ401
Lastname: Bwalya
Firstname: Bupe
Gender: F
School fees: 2 500.00

⚠ **Exam Tip:** Text/string values go in single quotes. Number values (like fees) do NOT use quotes. Make sure the number of fields matches the number of values.

14.4 DDL vs DML – SQL Classifications

Category	Keywords and Description
DDL – Data Definition Language	CREATE: Creates a new table or database. ALTER: Modifies the structure of an existing table. DROP: Deletes a table or database. TRUNCATE: Removes all rows from a table.
DML – Data Manipulation Language	INSERT: Adds new records to a table. SELECT: Retrieves data from a table. UPDATE: Modifies existing records. DELETE: Removes records from a table.

Exam Question: Classify the SQL keywords as DDL or DML: INSERT, ALTER, CREATE, UPDATE. [4]

Answer: DDL (Data Definition Language): ALTER, CREATE DML (Data Manipulation Language): INSERT, UPDATE

⚠ **Exam Tip:** Memory tip: DDL changes the STRUCTURE of the database (tables, columns). DML changes the DATA inside the tables.

Quick Reference Summary – Key Exam Answers

Q#	Question	Model Answer
Q1(a)	Multifunctional device example	All-in-One printer / Smartphone
Q1(b)	Primary memory example	RAM / ROM / Cache
Q1(c)	Two items read by MICR	Sort code AND Account number
Q2(a)	Two types of OS	Single-user / Multi-user (any two)
Q2(b)	Two user interfaces	GUI AND CLI
Q2(c)	Meaning of Scheduling	OS manages order/time each process uses the CPU
Q3(a)	Analogue vs Digital	Analogue = continuous wave; Digital = discrete 0/1 pulses
Q3(b)	Denary 50 to binary	110010
Q4(a)	Ring topology diagram	Devices in a circle, connected, labelled, arrows
Q4(b)	Server function	Stores/manages shared files; controls access
Q4(c)	Ring topology drawback	One failure breaks the whole network
Q5(a)	Logic circuit X=(A AND B) OR (B AND C)	Two AND gates → one OR gate
Q5(b)	Home appliance with logic circuits	Washing machine / Microwave
Q6(a)	Presentation application	Microsoft PowerPoint
Q6(b)	Presentation device	Projector
Q6(c)	Placeholder data type	Text / Image / Chart
Q7(b)	Benefit of computers in hospitals	Fast access to patient records
Q7(c)	Disadvantage in hospitals	Risk of data breach / hacking
Q8(a)	Two robot tasks in factory	Welding / Painting / Assembly
Q8(b)	Effect of robots on workers	Job losses / redundancy

Q#	Question	Model Answer
Q9(a)	Cell A3 data type	Text (Label)
Q9(c)	Formula for total learners	=SUM(B3:D3)
Q10(a)	Malware example	Virus / Worm / Trojan
Q10(b)	Two malware protection methods	Antivirus software; Keep OS updated
Q10(c)	Physical threat	Fire / Theft / Flooding
Q11(a)	Importance of Data Protection Act	Protects personal data from misuse
Q11(b)	Two DPA principles	Data must be secure; Data kept no longer than necessary
Q12	Pseudocode to flowchart	START→INPUT→DECISION→OUTPUT (login)→END
SB Q1(a)	Convert paper to electronic	Scanning + OCR / Manual data entry
SB Q1(b)	Advantage of computer system	Faster retrieval of information
SB Q1(b)	Drawback of computer system	High cost / risk of data loss
SB Q1(c)	Method to gather information	Interview / Observation / Questionnaire
SB Q1(d)	Two design stage activities	Design input forms; Design database structure
SB Q1(e)	Maintenance documentation	Technical/system documentation
SB Q1(f)	Two implementation methods	Direct changeover / Parallel running
SB Q2(a)	Use of #include<iostream>	Includes I/O library for cin and cout
SB Q2(b)	Errors in C++ program	Line 4: int main() { | Line 6: cin >> name;
SB Q2(c)	Meaning of keyword in C++	Reserved word with predefined meaning in the language
SB Q3(a)	Why Student_No is primary key	It uniquely identifies each student record
SB Q3(b)	Number of records	3
SB Q3(c)	SQL INSERT statement	INSERT INTO Students VALUES ('NJ401','Bwalya','Bupe','F',2500.00);
SB Q3(d)	DDL keywords	ALTER, CREATE
SB Q3(d)	DML keywords	INSERT, UPDATE

Good luck in your examinations!

These notes are based on the 2025 Computer Studies Grade 12 Paper 1 (ECZ/SC/2025/B5)